

Supplementary Material for: The Loyola Index Family: What a Degree-Imbalance Coordinate Can and Cannot Resolve

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This supplement accompanies the paper “The Loyola Index Family: What a Degree-Imbalance Coordinate Can and Cannot Resolve”. Section, table, figure, equation and proposition numbers without the prefix S refer to the main paper, and data, scripts and the verifier are available at <https://github.com/Singati2/LOYOLA-PAPER>.

S1 Octane property data: provenance audit and correlation heatmap

Table 5 of the main paper reports the Pearson correlation r between each index and each of the five physicochemical properties of the 18 octane isomers. The properties are the boiling point T_B , the enthalpy of formation ΔH_f , the enthalpy of vaporization ΔH_{vap} , the entropy S and the acentric factor ω . The property data were traced to the NIST WebBook [5], and

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for each value the provenance file records the WebBook entry or, where the WebBook lists no value, the compilation consulted. Acentric factors, which NIST does not list, are the exception and were taken from KDB-derived compilations. The same five properties form the octane benchmark used in [6]. T_B is in $^{\circ}\text{C}$, ΔH_f and ΔH_{vap} are in kcal mol^{-1} (the latter at 298 K), S is in $\text{cal mol}^{-1}\text{K}^{-1}$, and ω is dimensionless. Values for ΔH_{vap} and entropy S vary slightly across compilations, and all comparisons here are made within this single, internally consistent dataset. Calculations use this author-compiled property table attributed to the NIST WebBook. All 90 molecule-level values were traced against authoritative sources, with the NIST WebBook consulted first and KDB-derived compilations used for the acentric factor, which NIST does not list. The values were classified under a strict taxonomy. In it, “verified” requires agreement at the value’s reported precision, and a converted value exactly half a display unit away counts as agreement at reported precision. Of the values, 3 match exactly, and 29 match a specific recorded determination at reported precision after unit conversion, with the gas phase confirmed for formation enthalpies and entropies. A further 40 agree within the audit tolerances, the source’s stated uncertainty or the span of several determinations without matching any one of them at reported precision. 10 lie within explained variation among authoritative determinations, namely 1947 API Project 44 entries superseded by later TRC revisions. 7 could not be verified against a reachable authoritative value, six of these being gas-phase entropies absent from the NIST WebBook. For them the 1947 API Project 44 tables give values differing by 0.07–2.2 units, as they do for isomers whose later revisions were confirmed. The remaining one is the 298 K vaporization enthalpy of 2,2,3,3-tetramethylbutane. This compound is a solid at 298 K (triple point 374 K), and the NIST WebBook lists standard vaporization enthalpies of 42.94 and 42.91 kJ mol^{-1} (about 10.26 kcal mol^{-1}) together with sublimation enthalpies. The benchmark value used here, 8.41 kcal mol^{-1} (35.19 kJ mol^{-1}), is that of the standard octane benchmark compilation and matches no listed determination. Its physical reference state is therefore unresolved. Section S3 reports the ΔH_{vap} analysis with this isomer excluded. One value is a marginal con-

flict, the acentric factor of 2,2,3,3-tetramethylbutane (0.247 vs. compiled 0.251), while the acentric factor of 2,2,4-trimethylpentane (0.305 vs. compiled 0.303) agrees only within tolerance. Substituting the alternative compiled acentric factors changes several of the ω correlations in Table 5 of the main paper at the third decimal (e.g. M_2 moves from -0.988 to -0.986). It does not change the property-wise winning descriptor or any substantive interpretation, and the full sensitivity computation and the machine-readable provenance trace accompany the reproducibility package. For 3,3-dimethylhexane, 3-ethyl-3-methylpentane, 2,2,3-trimethylpentane and 2,3,3-trimethylpentane, values of ΔH_{vap} that differ from the NIST Web-Book by $0.05\text{--}0.13\text{ kcal mol}^{-1}$ circulate in octane benchmark tables. We use values consistent with NIST (Appendix A of the main paper). Two are exact conversions of NIST determinations (8.97, 9.08), and two lie within the span or stated uncertainty of the NIST determinations. These are 8.83, which also equals the API Project 44 value at the displayed precision, and 8.90, which equals the compiled octane dataset value 8.897 at the displayed precision. Six gas-phase entropies could not be verified against a primary source (see above). Four entropies (*n*-octane, 2,2-dimethylhexane, 2,2,4- and 2,3,3-trimethylpentane) differ from those of a widely used octane QSPR compilation by $0.04\text{--}2.29$ units, the 2,3,3-trimethylpentane value 102.06 being a compiled-dataset value (see Section S10). Substituting the compilation values changes all twelve S correlations in Table 5 of the main paper, by up to 0.043 in $|r|$, but it does not change the winning descriptor (HM). Apart from the four corrected vaporization enthalpies and the corrected transcription of the 2,3,3-trimethylpentane entropy (Section S10, item (ii)), the benchmark values are preserved as supplied for comparability.

Correlation details for Table 5 of the main paper. The best correlation per property is split among the reductions and the pure- γ points, with the modified second Zagreb index $LO(-1, 0, 0)$ strongest on T_B and ΔH_f . The hyper-Zagreb index $LO(0, 2, 0)$ is strongest on S , the second Zagreb index $LO(1, 0, 0)$ on ω , and the fixed pure- γ benchmark point $LO(0, 0, 1)$ on ΔH_{vap} ($|r| = 0.984$, with $LO(0, 0, 2)$ second at

0.981). At the exact-reduction triples the Loyola correlations coincide with the corresponding classical indices to floating-point precision, as they must. For each property, a descriptive post-selection paired bootstrap interval for the observed top-two $|r|$ difference included zero ($B = 10\,000$, resampling the 18 isomers jointly, `default_rng(1)`). These intervals were not multiplicity-adjusted across the five properties and do not establish equality or equivalence. Individual bootstrap half-widths range from 0.016 to 0.096 across the headline cells (percentile bootstrap, $B = 10\,000$, `default_rng(0)`). The value of $LO(0, 0, 1)$ on ΔH_{vap} is that of a fixed benchmark point and involves no tuning. Recomputing each bolded correlation with single isomers deleted shifts $|r|$ by at most 0.027, which indicates that no single isomer drives the leading correlations.

Descriptor redundancy. The descriptors are strongly collinear. This limits what any small $|r|$ difference in Table 5 of the main paper can mean. Across the twelve compared descriptors on the 18 isomers, fourteen descriptor pairs have $|r| \geq 0.99$, the extreme being R versus $H/2$ at $r = 0.9997$. A principal component analysis of the standardised 18×12 descriptor matrix places 90.3% of the variance on the first component and 99.05% on the first two. The effective rank (participation ratio of the eigenvalue spectrum) is 1.2, so the twelve-index comparison set is essentially a two-dimensional descriptor space. Differences in the third decimal of $|r|$ between near-collinear descriptors should not be read as meaningful rankings. Figure S1 shows the signed- r heatmap of the compared indices against the five properties, displaying the values of Table 5 of the main paper.

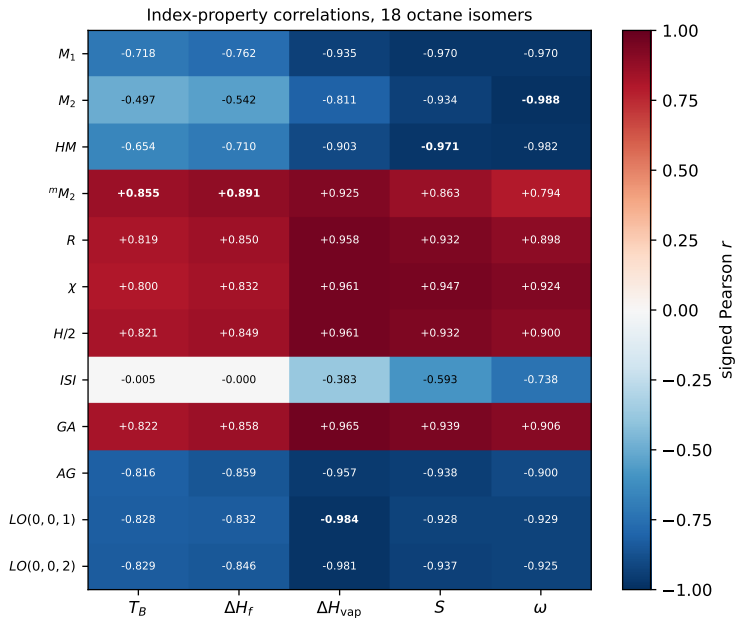


Figure S1. Signed Pearson r between the Loyola reductions, the fixed pure- γ benchmark points and the five octane physicochemical properties on the 18 isomers.

S2 Effect of parameter tuning near the best indices

The benchmark of Section 5.1 of the main paper tunes no parameters and compares only the exact reductions and the two fixed pure- γ benchmark points. It is nonetheless natural to ask whether small, data-driven adjustments of (α, β, γ) near the best-correlating indices could improve the fit, and this section examines that question as an exploratory check. We take the four anchors carrying the strongest column correlations in Table 5 of the main paper, namely $M_2 = LO(1, 0, 0)$, $HM = LO(0, 2, 0)$, $^mM_2 = LO(-1, 0, 0)$, and the pure- γ point $LO(0, 0, 1)$. For each anchor we draw 50 random parameter triples within ± 0.3 (to two decimal places) of the anchor and correlate each against the five octane properties. The

four pools are drawn, in the order listed, from one pseudo-random stream (NumPy `default_rng(42)`), each coordinate offset being a uniform draw on $[-0.3, 0.3]$ rounded to two decimals. The percentile bootstrap uses `default_rng(0)`. For every anchor–property cell, Table S1 reports the anchor correlation $|r|_0$, the best of the 50 perturbations in sample $|r|_{50}$, the in-sample gain Δ_{in} and the percentile-bootstrap half-width ($B = 10\,000$). It also reports the leave-one-out correlation of the anchor LOO_0 and a *nested* leave-one-out correlation LOO_{50} . In the nested version the best of the 50 candidates is selected, within each of the four globally selected anchor pools, using the 17 training isomers of each fold only. It is then evaluated on the held-out one. The resulting out-of-sample gain is Δ_{out} . The anchors themselves were chosen from the complete-data correlations of Table 5 of the main paper, and the 18 pooled fold predictions share heavily overlapping training sets. These values are therefore descriptive rather than independent replications.

Table S1. Decimal parameter tuning near the four best-correlating anchors on the 18 octane isomers (50 random triples within ± 0.3 of each anchor). The columns give the anchor correlation $|r|_0$, the best-of-50 in sample $|r|_{50}$, the in-sample gain Δ_{in} and the percentile-bootstrap half-width $\pm\text{boot}$ ($B = 10\,000$). They then give the leave-one-out correlation of the anchor LOO_0 , the foldwise nested leave-one-out correlation of the tuned family LOO_{50} , computed within the globally selected anchor pool, and the out-of-sample gain $\Delta_{\text{out}} = \text{LOO}_{50} - \text{LOO}_0$. Gains are computed from full-precision values and rounded once, so a displayed gain may differ by one unit in the last place from the difference of the displayed operands.

Property	$ r _0$	$ r _{50}$	Δ_{in}	$\pm\text{boot}$	LOO_0	LOO_{50}	Δ_{out}
<i>near</i> $M_2 = LO(1, 0, 0)$							
T_B	0.497	0.598	+0.100	0.353	0.331	0.458	+0.127
ΔH_f	0.542	0.635	+0.093	0.358	0.395	0.540	+0.145
ΔH_{vap}	0.811	0.876	+0.066	0.185	0.768	0.838	+0.071
S	0.934	0.955	+0.021	0.073	0.909	0.935	+0.025
ω	0.988	0.995	+0.007	0.018	0.985	0.985	0.000
<i>near</i> $HM = LO(0, 2, 0)$							
T_B	0.654	0.740	+0.085	0.257	0.545	0.647	+0.102
ΔH_f	0.710	0.786	+0.076	0.255	0.645	0.741	+0.097
ΔH_{vap}	0.903	0.944	+0.041	0.082	0.871	0.921	+0.049
S	0.971	0.973	+0.002	0.030	0.961	0.957	-0.004
ω	0.982	0.989	+0.007	0.025	0.979	0.982	+0.003
<i>near</i> ${}^mM_2 = LO(-1, 0, 0)$							
T_B	0.855	0.864	+0.009	0.096	0.781	0.759	-0.021
ΔH_f	0.891	0.897	+0.006	0.087	0.865	0.837	-0.028
ΔH_{vap}	0.925	0.966	+0.041	0.068	0.900	0.953	+0.053
S	0.863	0.917	+0.054	0.192	0.810	0.878	+0.068
ω	0.794	0.881	+0.088	0.246	0.730	0.853	+0.123
<i>near</i> $LO(0, 0, 1)$							
T_B	0.828	0.835	+0.007	0.161	0.771	0.764	-0.007
ΔH_f	0.832	0.842	+0.010	0.152	0.800	0.807	+0.006
ΔH_{vap}	0.984	0.985	+0.001	0.016	0.978	0.978	-0.001
S	0.928	0.945	+0.017	0.046	0.887	0.915	+0.028
ω	0.929	0.945	+0.016	0.074	0.907	0.926	+0.019

Three observations can be made. First, every in-sample gain Δ_{in} is smaller than the corresponding single-descriptor bootstrap uncertainty scale, although this comparison is descriptive rather than a formal paired or post-selection test of the gain. Second, under the foldwise nested evaluation (within the globally selected anchor pools) these descriptive point-estimate gains do not vanish. Tuning near M_2 raises the out-of-sample correlation for T_B and ΔH_f by +0.13 and +0.15, and tuning near mM_2 raises ω by +0.12. In every such case, however, the anchor was a *weak* descriptor for that property, and the tuned correlation only climbs part

way toward the level a strong reduction already attains. Tuning near M_2 lifts T_B to 0.46, whereas mM_2 already reaches 0.78. Where the anchor is already the column winner (M_2 on ω , HM on S , $LO(0,0,1)$ on ΔH_{vap}), the nested gain for this pool seed is -0.0003 , -0.004 and -0.001 . These near-zero values are specific to the seed. Re-drawing the pools with seeds 0–99, the nested gain for M_2 on ω is small but positive in most seeds (median $+0.003$, positive in 97 of 100 seeds). The sweep is in the repository script `tuning_pool_seed_sweep.py`. Tuning a column winner therefore changes its nested correlation by at most a few thousandths. Third, tuning near the pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ does not exceed the best fixed index under nested leave-one-out on any of the five properties (Table S2). On ΔH_{vap} the best fixed index is the pure- γ point $LO(0,0,1)$ itself. There the tuned value exceeds it by 0.001 in sample (0.985 vs 0.984) and falls 0.001 below it under leave-one-out (both 0.978 at three decimals, the differences being those of full-precision values). These post-selection differences are negligible. On T_B , ΔH_f , S , and ω the best fixed reduction is higher both in and out of sample.

Table S2. Decimal tuning near the pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ versus the best fixed index in the tested comparison set for each octane property. The best fixed index is ranked by in-sample $|r|$ (its leave-one-out value in parentheses). The decimal columns give the best over two pools of 50 random triples within ± 0.3 of $LO(0,0,1)$ and of $LO(0,0,2)$, in sample and under foldwise nested leave-one-out within the globally selected pools. Here the $LO(0,0,1)$ pool is exactly the pool of Table S1, and the $LO(0,0,2)$ pool is the next 50 triples of the same `default_rng(42)` stream. Indices are ranked by in-sample $|r|$ rather than LOO, as the latter is inflated by degenerate fits for near-zero-correlation indices. Tuning does not exceed the best fixed index under LOO on any property. In sample it exceeds it only on ΔH_{vap} , by 0.001, a negligible post-selection difference that does not establish superiority.

Property	best fixed	$ r _{\text{fix}}$	LOO_{fix}	$ r _{\text{dec}}$	LOO_{dec}
T_B	mM_2	0.855	0.781	0.835	0.764
ΔH_f	mM_2	0.891	0.865	0.850	0.815
ΔH_{vap}	$LO(0,0,1)$	0.984	0.978	0.985	0.978
S	HM	0.971	0.961	0.945	0.915
ω	M_2	0.988	0.985	0.945	0.926

In short, local tuning raised several nested leave-one-out point estimates for weak anchors. Except for the 0.001 in-sample ΔH_{vap} margin noted above, it did not exceed the best fixed benchmark in the tested comparison set. No formal superiority claim is made. All tabled nested values correspond to positive signed prediction–observation correlations with positive predictive Q^2 . A constant-descriptor control confirms that absolute pooled correlation alone can be inflated for degenerate descriptors (cf. the caveat in the caption of Table S2).

S3 Matched-budget comparison in the search box $(-2, 2)$

The analyses above tune around fixed anchors and are descriptive. The question raised by the Gutman–Milovanović framing is whether, given identical search budgets, freeing γ improves out-of-sample prediction beyond an optimised parent $M_{\alpha,\beta}$. This is tested with a fully nested leave-one-out design. For each property and each outer fold (one isomer held out), 200 candidate parameter vectors are drawn once from Uniform $(-2, 2)$ per coordinate. The candidates are triples (α, β, γ) for the Loyola model and pairs (α, β) with $\gamma = 0$ for the parent. The budget is identical. Candidates are rounded to three decimals and drawn from NumPy `default_rng(12345)` for Table S3, Loyola triples first. On the 17 training isomers only, every candidate is scored by *inner* leave-one-out root-mean-square error of the one-descriptor linear regression. The best candidate is selected, the regression is refit on all 17, and the held-out isomer is predicted. No quantity computed on the held-out isomer enters selection or fitting at any point.

Once the parent is searched over a box that does not truncate it, freeing γ gives no consistent gain (Table 5 and Section 5.2 of the main paper). The comparison in the box $(-2, 2)$ is reported first because it shows how the apparent gain arose, and in that box the result is mixed (Table S3). For the illustrative seed, freeing γ improves nested predictive performance on two of the five properties (T_B , $\Delta Q^2 = +0.136$, and ΔH_{vap} , $\Delta Q^2 = +0.086$)

Table S3. Matched-budget nested leave-one-out comparison of the optimised parent $M_{\alpha,\beta}$ (GM) and the optimised Loyola family (γ free) on the five octane properties. Panel (a) is a *single illustrative candidate stream* (seed 12345, budget 200), where Q^2 is the outer predictive coefficient and RMSE the outer root-mean-square error. “LO better” counts outer folds (of 18) in which the Loyola model had the strictly smaller absolute error. Panel (b) summarises the 100-seed sweep (seeds 0–99), giving the median ΔQ^2 with its [minimum, maximum], and the number of seeds in which $\Delta Q^2 > 0$, at budgets $B = 200$ and $B = 500$. All selection is repeated inside every outer fold, and all values are full-precision results rounded once.

<i>(a) single candidate stream: seed 12345, budget 200</i>						
Property	Q_{GM}^2	Q_{LO}^2	ΔQ^2	RMSE _{GM}	RMSE _{LO}	LO better
T_B	+0.613	+0.748	+0.136	3.700	2.983	10/18
ΔH_f	+0.855	+0.549	−0.306	0.463	0.816	4/18
ΔH_{vap}	+0.884	+0.971	+0.086	0.127	0.064	13/18
S	+0.906	+0.878	−0.029	1.363	1.558	7/18
ω	+0.979	+0.956	−0.022	0.005	0.008	5/18

<i>(b) 100 candidate streams: seeds 0–99</i>				
Property	median ΔQ^2 [min, max]		seeds with $\Delta Q^2 > 0$	
	$B = 200$	$B = 500$	$B = 200$	$B = 500$
T_B	+0.010 [−0.524, +0.373]	+0.066 [−0.307, +0.231]	53/100	64/100
ΔH_f	−0.120 [−0.408, +0.206]	−0.095 [−0.373, +0.118]	14/100	24/100
ΔH_{vap}	+0.069 [+0.034, +0.102]	+0.068 [+0.024, +0.102]	100/100	100/100
S	−0.012 [−0.140, +0.038]	−0.010 [−0.098, +0.031]	34/100	31/100
ω	−0.010 [−0.074, +0.016]	−0.006 [−0.052, +0.012]	26/100	28/100

and degrades it on three (ΔH_f , S , ω). The ΔH_f degradation is driven by selection instability. The parent model selected the same (α, β) in all 18 folds there, while the three-parameter search selected widely varying triples (fold-to-fold standard deviation above 1.2 in both α and β). Selected γ values are interior to the search box (maximum $|\gamma| = 0.48$). The parent’s selections are not. For ΔH_{vap} the parent chose $(\alpha, \beta) = (1.021, -1.993)$, on the boundary of the box, in 16 of the 18 folds, a choice whose consequences are examined in Section 5.2 of the main paper. On the one property where γ helps most stably, ΔH_{vap} , the identical triple $(-0.270, +0.271, +0.239)$ was selected in every fold. With $n = 18$ molecules these Q^2 differences carry substantial selection noise. We attach no inferential claim to them.

Because the comparison depends on a random candidate stream, the full nested design was repeated for one hundred seeds (0–99) at both budgets (200 and 500 candidates per model). This makes 1000 configurations in all. A paired control was run alongside, comparing the Loyola candidates against the *same* (α, β) coordinates with γ set to zero, which isolates the effect of γ from the independent candidate streams. Freeing γ improved nested predictive performance for ΔH_{vap} in every seed of this sweep (100/100 at both budgets, and 100/100 in the paired control). The median was $\Delta Q^2 = +0.069$ and $+0.068$, with range $+0.024$ to $+0.102$. It *usually* degraded performance for ΔH_f (86/100 and 76/100 seeds), while for T_B the split is close to even at budget 200 (53/100) but leans towards improvement at budget 500 (64/100). S and ω usually degrade slightly. These counts describe optimiser variance over random candidate streams on one fixed set of 18 molecules, which are never re-sampled. The counts therefore carry no inferential weight. In particular, 100/100 does not establish that γ improves ΔH_{vap} prediction for other molecules or data sets. The result is also sensitive to a single data cell, the ΔH_{vap} value of 2,2,3,3-tetramethylbutane, whose physical reference state is unresolved (Section S1). The complete nested comparison was repeated with that isomer excluded (17 molecules, same seeds, budgets and candidate streams). Freeing γ then improved ΔH_{vap} in 93/100 seeds at budget 200 and 96/100 at budget 500 (paired control 95/100 and 96/100). The median was $\Delta Q^2 = +0.046$ and $+0.051$, with range -0.048 to $+0.098$.

The gain survives the exclusion but is no longer sign-invariant. These comparisons share the box $(-2, 2)$, which truncates the parent along the direction $\beta \approx -2\alpha$. Section 5.2 and Table 6 of the main paper show that the vaporization-enthalpy gain shrinks to about zero once the parent is searched over a box that does not truncate it.

S4 Nonane replication test: protocol, exploratory transfer and molecule bootstrap

Two limitations of the octane analysis are its size (18 molecules) and its restriction to index models. Both were addressed under a protocol recorded in a git commit of the reproducibility package before any nonane value was collected. The identical nested design was applied to the 35 constitutional nonane isomers. Boiling points (35 values) and standard vaporization enthalpies (34 values, none being listed for 3-ethyl-4-methylhexane) were taken from the NIST WebBook [5], with every value traced to its archived source page. The two properties differ in evidential status. Twenty of the 35 boiling points are NIST averages of experimental determinations (13 quoted only to 1 K). Five are single NIST determinations and ten are the most recent primary determination listed by NIST (Deviation D1 of the protocol). All are experimental measurements, so the T_B test is experimentally grounded. Of the vaporization enthalpies, only 4 of 34 are calorimetric determinations and one is a NIST average. The other 29 come from two compilations. These are the 1961 tabulation of Labauf, Green-shields and Rossini [4], which covers all 35 nonanes and 75 decanes and is not a set of individual calorimetric measurements, and a 1971 vapour-pressure handbook [8]. Such tabulations may rest partly on structural-increment estimates, so agreement with them measures consistency with the compiled values rather than independent experimental validation. A degree-pair-based predictor may also be favoured if the compiled values were themselves derived from structural increments. The pre-registered decision rule called the γ effect *replicated* if the Loyola model beat the

parent in at least 80 of 100 seeds at both budgets with positive median ΔQ^2 . Four baselines were added, each fitted entirely within the outer training fold. They are the training-fold mean, the best of the ten classical indices of Table 5 of the main paper, and the irregularity index *IRLA*. The fourth is ridge regression on the full edge-degree-pair count vector, with its penalty chosen by inner leave-one-out, and the resulting Q^2 values are in Table 6 of the main paper.

On the nonane replication set the pre-registered verdicts are as follows. For ΔH_{vap} (compilation values) the γ effect met the pre-registered optimiser-stability criterion and *replicates*, LO being better than GM in 95/100 and 98/100 seeds, with median $\Delta Q^2 = +0.048$ and $+0.053$. For T_B the verdict is *mixed* (58/100 and 63/100). It is unchanged in direction under the pre-registered sensitivity analysis D4 of the protocol. That analysis replaces the NIST average for 2,3,3,4-tetramethylpentane, 413 ± 6 K, by the median of all eight listed determinations including a 1989 handbook value and a 1939 outlier, 414.62 K. LO is then better in 60/100 and 66/100 seeds, with median $\Delta Q^2 = +0.041$ and $+0.052$. The pre-registered protocol fixed the box of half-width 2, which, as Table 6 of the main paper shows, binds the parent. In a box that does not, the nonane gain disappears (median $\Delta Q^2 = -0.002$ under the hybrid control), and the pre-registered verdict is reported as obtained, with this deviation recorded. The pattern behind the vaporization result is the same on both sets. Models whose edge weight depends on imbalance only through q^2 inside the box of half-width 2 (the classical indices, including $GA = \sum_{uv} \sqrt{1 - q_{uv}^2}$, and GM) reach $Q^2 = 0.853$ – 0.888 . Models with a term linear in q or with unrestricted counts (*IRLA*, ridge, $LO(0, 0, 1)$, LO) reach $Q^2 = 0.897$ – 0.957 . This split holds only inside the narrow box. In the box of half-width 12 the parent alone reaches 0.942 and 0.892 (main paper, Table 5). The fixed point $LO(0, 0, 1)$ involves no tuning at all, and it matches or beats the tuned Loyola model on both sets (0.957 and 0.903 against medians 0.953 and 0.900). It is also the index selected in every fold for ΔH_{vap} when admitted to the fixed-index comparison. It is closely related to *IRLA*. One has $LO(G; 0, 0, \gamma) = m + \frac{\gamma}{2} IRLA(G) + O(\gamma^2)$ as $\gamma \rightarrow 0$, and on both sets the correlation between $LO(0, 0, 1)$ and *IRLA* is 0.996. The base-

lines, however, also show the limit of the claim. Ridge regression on the full degree-pair count vector predicts the nonanes better than every index model, for both properties ($Q^2 = 0.927$ for ΔH_{vap} , 0.682 for T_B). On the octanes it matches the Loyola model on ΔH_{vap} and is weaker elsewhere. This is consistent with Proposition 6 of the main paper and the remark after Proposition 7. The imbalance coordinate re-weights degree-pair information rather than adding to it, and one interpretable descriptor recovers most of what a model on all ten degree-pair counts achieves for the vaporization enthalpy. It does not outperform such a model. The specific exponential form is not essential either, since the parameter-free aggregate *IRLA* comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900). For this property a term linear in the imbalance accounts for the effect.

Finally, the nested design refits every model on the nonanes. It therefore tests whether the *procedure* replicates rather than whether an octane-trained model transfers. An exploratory analysis followed. It was not pre-registered, and it was run in the box of half-width 2, so that it inherits the truncation of the parent. In it, the parameters selected on all 18 octanes were frozen and applied to the nonanes. With the parameters frozen and only the linear calibration refitted, the Loyola model beat the parent for ΔH_{vap} in 100/100 seeds at both budgets (median $Q^2 = 0.910$ and 0.916 against 0.864 and 0.865). It beat the parent for T_B in 65/100 and 82/100. With the octane calibration frozen as well, both models extrapolate poorly from C_8 to C_9 . The Loyola model is then worse than the parent (median Q^2 between -2.22 and -0.72 for LO against 0.06 to 0.44 for GM). Because both comparisons inherit the truncated box, they do not bear on whether γ adds predictive information.

The seed counts above measure optimiser variance only. To indicate molecule-level uncertainty we also computed, post hoc, a paired bootstrap over molecules (2000 resamples) of the median-over-seeds ΔQ^2 , conditional on the fitted outer predictions. Selection is not repeated within resamples. The intervals therefore do not account for uncertainty from refitting and model selection. The 95% intervals for LO minus GM on ΔH_{vap} are $[+0.014, +0.111]$ on the octanes but $[-0.006, +0.113]$ on the nonanes at

budget 200 ($[-0.005, +0.119]$ at 500). Although the nonane gain met the pre-registered criterion, its interval narrowly includes zero, as does the interval for LO minus ridge on the nonanes, $[-0.101, +0.032]$. For T_B the intervals are wide ($[-0.212, +0.222]$ for LO minus GM on the nonanes). The one octane interval that excludes zero in the other direction is that for ΔH_f , where freeing γ is worse ($[-0.281, -0.044]$). All these intervals refer to the box of half-width 2. Given Table 6 of the main paper, no γ effect remains to be established. The intervals are reported for completeness.

S5 Degeneracy across tree orders

Degeneracy [3] measures the failure of an index to discriminate non-isomorphic graphs, so lower values are better. On the 106 non-isomorphic trees of order 10 we use the distinct-value convention

$$\text{Degeneracy}(\%) = 100 \left(1 - \frac{k}{N} \right), \quad k = \#\{\text{distinct index values}\}, \quad N = 106,$$

i.e. the number $N - k$ of structures in excess of the distinct values, as a percentage of N . This is not the share of structures lacking a unique value, which is larger. For $LO(0, 0, 1)$ on this set, 45 of the 106 trees (42.5%) share their value with another tree. The pure- γ points $LO(0, 0, 1)$ and $LO(0, 0, 2)$ attain degeneracy 25.5% (79 distinct values), among the lowest of all compared indices and tied with R , GA and AG . The 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the minimum degeneracy attainable by *any* degree-pair-based index on this set, and the pure- γ points attain this floor. Theorem 2 of the main paper gives the reason and the order at which this stops. The first Zagreb index $LO(0, 1, 0)$ (83.0%) and the second Zagreb index $LO(1, 0, 0)$ (67.0%) are the most degenerate (Figure S2).

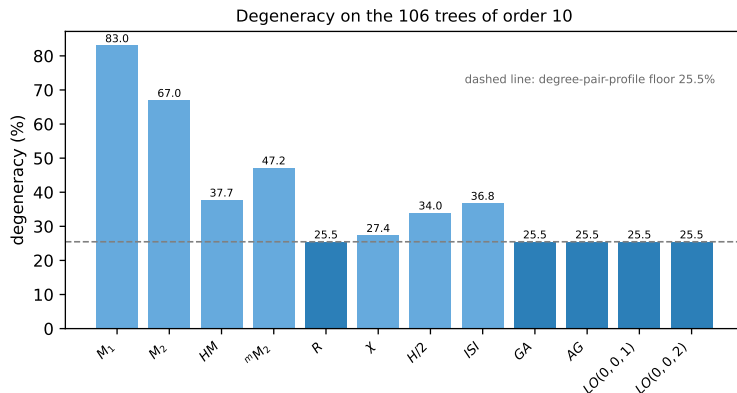


Figure S2. Degeneracy percentage of the compared indices on the 106 non-isomorphic trees of order 10. The dashed line marks the degree-pair-profile floor, $25.5\% = 100(1 - 79/106)$. The 106 trees realise only 79 distinct edge-degree-pair count vectors, so no vertex-degree-based index can fall below it.

To test whether this floor attainment is peculiar to order 10, the same computation was repeated for all non-isomorphic trees of orders 7 through 12. It was also repeated, separately, for the molecular subsets with $\Delta \leq 4$ (Table S4), and the pure- γ points attain the degree-pair-profile floor at *every* order in both sets. The Randić index also attains the floor at every order in this range, while χ falls off the floor from order 10. Floor attainment therefore distinguishes the pure- γ points from the most degenerate classical indices (M_1 , M_2) but not from R . The imbalance weights therefore lose nothing, although they are not unique in this respect. Theorem 2 of the main paper turns this into an exact statement. The pure- γ points separate precisely the q -histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for $\Delta \leq 4$) and not beyond. Among the trees of order 13 the profiles $\{(1, 2)^1, (1, 3)^2, (1, 6)^5, (2, 3)^2, (2, 6)^1, (3, 3)^1\}$ and $\{(1, 3)^3, (1, 6)^5, (2, 2)^1, (2, 3)^2, (3, 6)^1\}$ (exponents are edge multiplicities) share the q -histogram $\{0^1, \frac{1}{5}^2, \frac{1}{3}^1, \frac{1}{2}^3, \frac{5}{7}^5\}$, and there the pure- γ points realise 566 distinct values against 570 profiles. Among molecular trees ($\Delta \leq 4$) the first failure occurs at order 16 (809 values against 810 profiles). The Randić index still attains the floor at order 13 (570 values)

and falls below it at order 14 (1095 values against 1100 profiles, where the pure- γ points give 1090).

Table S4. Discrimination across tree orders $n = 7$ –12. The columns give the number of trees N , the number of distinct edge-degree-pair profiles (the floor attainable by any vertex-degree-based index), and distinct-value counts for representative indices. “mol.” denotes the molecular subset $\Delta \leq 4$.

n	set	N	floor	M_1	M_2	R	$LO(0, 0, 1)$	$LO(0, 0, 2)$
7	all	11	11	7	9	11	11	11
7	mol.	9	9	5	7	9	9	9
8	all	23	21	9	17	21	21	21
8	mol.	18	16	6	13	16	16	16
9	all	47	40	13	22	40	40	40
9	mol.	35	28	7	16	28	28	28
10	all	106	79	18	35	79	79	79
10	mol.	75	49	8	21	49	49	49
11	all	235	152	21	43	152	152	152
11	mol.	159	83	9	25	83	83	83
12	all	551	294	27	57	294	294	294
12	mol.	355	137	10	28	137	137	137

Fixed fractional exponents (exploratory). A script in the repository (directory `structural`) evaluates the nine fixed points

$$(\alpha, \beta) \in \left\{ (0, \tfrac{1}{2}), (\tfrac{1}{2}, 0), (0, -\tfrac{1}{2}), (-\tfrac{1}{2}, 0), (\tfrac{1}{2}, -1), \right. \\ \left. (-\tfrac{1}{2}, 1), (\tfrac{1}{3}, 0), (0, \tfrac{1}{3}), (\tfrac{1}{6}, 0) \right\}$$

with $\gamma = 0$ and $\gamma = 1$. We first consider discrimination on all trees of orders 12–16, where every point off the ratio plane with $\gamma = 1$ reaches or nearly reaches the profile floor. At order 16, for example, the counts are 4069, 4067, 4069, 4068, 4069 and 4069 of 4069 for $(0, \frac{1}{2})$, $(\frac{1}{2}, 0)$, $(0, -\frac{1}{2})$, $(-\frac{1}{2}, 0)$, $(\frac{1}{3}, 0)$, $(0, \frac{1}{3})$. With $\gamma = 0$ the same points give 3475, 3940, 3476, 4007, 4009 and 3476, while the two points on the ratio plane give 3996 for both values of γ . We next consider prediction (fixed points, leave-one-out Q^2 , no tuning). Here $\gamma = 1$ beats $\gamma = 0$ in 52 of 81 point–dataset combinations, mostly for the vaporization enthalpy and decane boiling points, and loses mostly on the octane entropy and acentric factor. Every

fixed fractional point stays at $Q^2 \leq 0.21$ on the decanes. The full output is committed with the script.

S6 Structure sensitivity: protocol and published control

The structure-sensitivity criterion for descriptor quality was introduced by Furtula, Gutman and Dehmer [2], and a fingerprint-based variant is due to Rakić and Furtula [7]. We implement the published protocol on the 75 decane isomer trees Ψ . For each tree G , the similar-structure set $S(G) \subset \Psi$ consists of the non-isomorphic trees obtainable from G by relocating a single edge within the molecular class (graph edit distance two, $\Delta \leq 4$ preserved). Then

$$SS(TI, G) = \frac{1}{|S(G)|} \sum_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

$$Abr(TI, G) = \max_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

and the class-level $SS(TI)$ and $Abr(TI)$ are the means over all 75 trees. On this class every $S(G)$ is nonempty (sizes 7 to 43). The neighbour relation is symmetric, and $Abr \geq SS$ holds throughout, as it must. As an external control, the implementation reproduces the published decane-class values of Barman and Das [1] for the seven indices common to both studies (M_1 , mM_2 , R , χ , H , GA , AG). These give fourteen tabulated SS/Abr values, of which eleven agree exactly at the four displayed decimals and the remaining three differ by one unit in the fourth decimal. The cell-by-cell comparison is in the file `fgd_published_control.csv` of the reproducibility package, while our values for all twelve compared indices are in Table 3 of the main paper. The published desideratum is SS large and Abr small [2]. Under it, the pure- γ point $LO(0, 0, 2)$ has a structure sensitivity equal to that of the hyper-Zagreb index (0.1257 against 0.1256) at a markedly smaller abruptness (0.274 against 0.313). The sensitivity-to-abruptness ratios of the pure- γ points (0.4579 and 0.4565) are comparable

to those of $H/2$, GA and R (0.448–0.451). In view of the redundancy analysis of Section 5.1 of the main paper and Section S1, these third-decimal differences are not a ranking. Two qualifications limit this. First, SS is not a fixed property of the family, since along the pure- γ line it grows with γ (0.031, 0.062, 0.126, 0.319 at $\gamma = 0.5, 1, 2, 5$). The ratio peaks near $\gamma = 2$ and then falls (0.427 at $\gamma = 5$). Second, the parameter-free aggregate $IRLA$, which the pure- γ points approximate to first order in γ , has a higher ratio ($SS = 0.2343$, $Abr = 0.5066$, ratio 0.4625). The favourable sensitivity–abruptness balance is therefore a property of linear imbalance weighting rather than of the exponential form.

S7 Lean 4 formalisation: scope and encoding

The directory `formal/` of the repository contains a Lean 4 formalisation (Lean v4.34.1, Mathlib v4.34.1) of the following statements of the main paper. `formal/verify.sh` builds it and checks that no `sorry`, `axiom` or `native_decide` occurs, and that every theorem depends only on the standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

Encoding. Graphs are not formalised as Mathlib graphs, and an edge set is a finite family of edges with a map to (positive real) degree pairs, or a multiset of degree pairs. Hypotheses such as “maximum degree Δ ” are stated as bounds on these pairs, so the link “these are the vertex degrees of a graph” is not formalised. The only exception is the two order-13 trees below, which are realised by explicit parent arrays.

Formalized (inequalities and identities, with their hypotheses). The formalised statements are the upper and lower (Δ, δ) bounds, the Cauchy–Schwarz interpolation bound and the ratio bound (and the bracket form). They also include the convexity of $\theta \mapsto \log LO(G; \theta)$ for any nonempty edge set (via Hölder), and the spanning statement of the independence lemma for every $\Delta \geq 3$ (via $\det W = \frac{1}{6} \ln 2 (8 \ln 3 - 13 \ln 2) \neq 0$, using $3^8 \neq 2^{13}$). Finally, the identity $1 - q^2 = 4xy/(x + y)^2$ is formalised, as is the order-13 profile pair with equal q -histograms and hence equal pure- γ values for every real γ , together with explicit 13-vertex trees realising both profiles.

Not formalised. The equality characterisations are not formalised, and neither are the gradient and Hessian formulas and strict convexity. The ratio-plane theorem, the γ -crossing bound and the identifiability proposition are not formalised either. These rely on the Lindemann–Weierstrass theorem, Laguerre’s rule and the theory of exponential families, which are not available in Mathlib. All statistical and chemometric results are outside the formalisation, as are the closed forms and the ceiling and generic-discrimination propositions. The identifiability consequence of Lemma 1 together with its $\Delta = 1, 2$ cases is also not formalised. A theorem-by-theorem map is in `formal/README.md`.

S8 Further directions

The following open problems remain:

1. Sharp two-sided bounds for $LO(G; \alpha, \beta, \gamma)$ via degree-based indices lying outside the family, such as the atom-bond-connectivity index ABC and the Sombor index.
2. Extremal-graph characterisations (not attempted in this paper) for several graph classes, including trees, unicyclic and bicyclic graphs.
3. Chemometric extension to additional chemical compound classes such as benzene derivatives, alkane homologues beyond the octanes, nonanes and decanes studied here and to compound-class-specific QSPR endpoints.

The first orders at which the pure- γ points, the Randić index and GA lose discrimination are summarised in Table 2 of the main paper.

S9 Closed forms, the log-convexity proof and termwise bounds

This section contains closed-form values on standard graphs, the proof of the global log-convexity theorem (Theorem 1 of the main paper), and termwise bounds for the family.

Closed-form values on standard graphs

The classical graph families used are the complete graph K_n , the cycle C_n , the path P_n , the star S_n and the complete bipartite graph $K_{p,q}$. Beyond these, the hypercube Q_k has 2^k vertices and $k \cdot 2^{k-1}$ edges and is k -regular. The wheel W_n is an $(n-1)$ -cycle plus a universal vertex, and the friendship graph F_p consists of p triangles sharing a common vertex. Finally, the r -regular graphs are also used.

Proposition 1. *Let $\alpha, \beta, \gamma \in \mathbb{R}$. Then:*

1. $LO(K_n; \alpha, \beta, \gamma) = \frac{n(n-1)}{2} 2^\beta (n-1)^{2\alpha+\beta}$ for $n \geq 2$.

2. $LO(C_n; \alpha, \beta, \gamma) = n \cdot 4^{\alpha+\beta}$ for $n \geq 3$.

3. $LO(P_n; \alpha, \beta, \gamma) = \begin{cases} 2^\beta, & n = 2, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3}, & n = 3, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3} + (n-3) \cdot 4^{\alpha+\beta}, & n \geq 4. \end{cases}$

4. $LO(S_n; \alpha, \beta, \gamma) = (n-1)(n-1)^\alpha n^\beta \exp\left(\gamma \frac{n-2}{n}\right)$ for $n \geq 2$.

5. $LO(K_{p,q}; \alpha, \beta, \gamma) = pq(pq)^\alpha (p+q)^\beta \exp\left(\gamma \frac{|p-q|}{p+q}\right)$.

6. $LO(Q_k; \alpha, \beta, \gamma) = k \cdot 2^{k-1} \cdot 2^\beta k^{2\alpha+\beta}$ for $k \geq 1$.

7. For $n \geq 5$, W_n has $n-1$ spoke edges with degree pair $(3, n-1)$ and $n-1$ rim edges with degree pair $(3, 3)$, so

$$LO(W_n; \alpha, \beta, \gamma) = (n-1) \left[3^\alpha (n-1)^\alpha (n+2)^\beta \exp\left(\gamma \frac{n-4}{n+2}\right) + 9^\alpha 6^\beta \right].$$

8. For $p \geq 1$, F_p has $2p$ spoke edges $(2, 2p)$ and p outer edges $(2, 2)$, so

$$LO(F_p; \alpha, \beta, \gamma) = p \cdot 4^{\alpha+\beta} + 2p(4p)^\alpha (2p+2)^\beta \exp\left(\gamma \frac{p-1}{p+1}\right).$$

9. For an r -regular graph G with m edges, $LO(G; \alpha, \beta, \gamma) = m \cdot 2^\beta r^{2\alpha+\beta}$.

Proof. Each case follows by enumerating the edge degree pairs and applying the definition of the main paper. In K_n , C_n , Q_k and r -regular graphs

every edge has $d_u = d_v = r$, so $\text{imb} = 0$ and $(d_u d_v)^\alpha (d_u + d_v)^\beta = 2^\beta r^{2\alpha+\beta}$, which gives items (1), (2), (6) and (9). All $n - 1$ edges of S_n have degree pair $(1, n - 1)$ with $\text{imb} = (n - 2)/n$ (item (4)), and all pq edges of $K_{p,q}$ have pair (p, q) (item (5)). For $n \geq 4$, P_n has two pendant edges $(1, 2)$ and $n - 3$ interior edges $(2, 2)$, and the cases $n = 2, 3$ are immediate (item (3)). Items (7) and (8) follow by summing over the two edge-degree-pair classes of W_n and F_p . ■

Proof of the global log-convexity theorem

Proof of Theorem 1 of the main paper. Direct differentiation of $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$ gives $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$ and $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k] \mathbb{E}_\theta[z_l]$, the (k, l) -entry of $\text{Cov}_\theta(z)$, which is positive semidefinite. Convexity follows. The measure $w(\theta)$ has full support on E for every θ . Hence $\text{Cov}_\theta(z)$ is singular in a direction v iff $v^\top z_{uv}$ is constant over all edges, i.e. iff the z_{uv} lie in a proper affine subspace, which proves the strictness criterion. Since $q_{uv} \in [0, 1)$, $\mathbb{E}_\theta[q] \in [0, 1)$, and $\mathbb{E}_\theta[q] = 0$ iff $q_{uv} = 0$ on every edge. On an r -regular graph all z_{uv} equal $z_0 = (2 \ln r, \ln 2r, 0)$. ■

Termwise bounds

We give a bracket bound via the edge-contribution function ψ , bounds in (m, Δ, δ) for non-negative parameters, a Cauchy–Schwarz interpolation bound and a uniform ratio bound. The $\gamma \neq 0$ bounds rest on bounding $e^{\gamma q_{uv}}$ edge by edge (for $\gamma \geq 0$) between 1 and $e^{\gamma(\Delta-1)/(\Delta+1)}$. Throughout, G is a simple connected graph with $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$. Several of the inequalities have been machine-checked (Section S7).

Remark 1 (Attribution at $\gamma = 0$). At $\gamma = 0$ the family coincides with $M_{\alpha,\beta}$, and inequalities of all the types developed here are already known for it. These include (Δ, δ, m) -bounds with regular/biregular equality, two-sided comparisons against M_1 , M_2 , R , χ , H , GA , AG and ISI [10], Jensen- and Hölder-type relations [9] and further inequalities [11]. The $\gamma = 0$ specialisations below are therefore not claimed as new. What this section adds is their extension to $\gamma \neq 0$, where the imbalance factor $e^{\gamma q_{uv}}$ enters

every edge weight, and the tracking of how γ deforms the equality cases. Comparisons at $\gamma = 0$ with the Randić, sum-connectivity and Sombor indices are not repeated.

The summand in the defining sum of the main paper depends only on the unordered edge degree pair (d_u, d_v) . We isolate this dependence as follows:

Definition 1. For real parameters α, β, γ and positive reals x, y ,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^\alpha (x + y)^\beta \exp\left(\gamma \frac{|x - y|}{x + y}\right). \quad (\text{S1})$$

With this notation $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$.

Proposition 2. *Let G be a simple connected graph of size $m \geq 1$ with degrees in $[\delta, \Delta]$. Then*

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \quad (\text{S2})$$

where, $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ and $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$. The upper inequality is an equality if and only if every edge of G has a degree pair attaining $\psi_{\max}$. The lower inequality is an equality if and only if every edge has a degree pair attaining $\psi_{\min}$.

Proof. Every edge uv has its degree pair in the admissible set $\mathcal{P}(\delta, \Delta) := \{(i, j) \in \mathbb{Z}^2 : \delta \leq i \leq j \leq \Delta\}$, hence $\psi_{\min} \leq \psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}$ on every edge, and summing over the m edges and using $\sum_{uv} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$ gives (S2). Each side of (S2) is the sum of m per-edge inequalities, and is therefore an equality if and only if every per-edge inequality is. That is, equality holds if and only if every edge has a degree pair attaining $\psi_{\max}$ (upper side), respectively $\psi_{\min}$ (lower side), over $\mathcal{P}(\delta, \Delta)$. ■

Corollary 3. *Let $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, and $\gamma_1 + \gamma_2 = 2\gamma$. Then for every simple connected graph G without isolated vertices,*

$$LO(G; \alpha, \beta, \gamma) \leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) \cdot LO(G; \alpha_2, \beta_2, \gamma_2)}$$

Equality holds if and only if the ratio

$$\psi(d_u, d_v; \alpha_1, \beta_1, \gamma_1) / \psi(d_u, d_v; \alpha_2, \beta_2, \gamma_2)$$

is the same for every edge of G .

Proof. For each edge $e = uv$ write $\psi_e^{(k)} := \psi(d_u, d_v; \alpha_k, \beta_k, \gamma_k) > 0$, $k \in \{1, 2\}$, and by additivity of the exponents and the hypotheses on $(\alpha_k, \beta_k, \gamma_k)$, $\psi_e^{(1)} \psi_e^{(2)} = \psi(d_u, d_v; \alpha, \beta, \gamma)^2$ on every edge. The Cauchy–Schwarz inequality for $a_e = (\psi_e^{(1)})^{1/2}$ and $b_e = (\psi_e^{(2)})^{1/2}$ gives $\sum_e a_e b_e \leq (\sum_e a_e^2)^{1/2} (\sum_e b_e^2)^{1/2}$, which is the claim because $\sum_e a_e b_e = LO(G; \alpha, \beta, \gamma)$, $\sum_e a_e^2 = LO(G; \alpha_1, \beta_1, \gamma_1)$ and $\sum_e b_e^2 = LO(G; \alpha_2, \beta_2, \gamma_2)$. Equality holds if and only if $b_e = \lambda a_e$ for a constant $\lambda \geq 0$ and all e , i.e. if and only if $\psi_e^{(2)}/\psi_e^{(1)} = \lambda^2$ is the same on every edge. (The inequality itself is midpoint log-convexity of $\theta \mapsto LO(G; \theta)$, Theorem 1 of the main paper.) ■

Proposition 4. *Let G be a connected graph of size m and maximum degree Δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \leq m \cdot \Delta^{2\alpha} \cdot (2\Delta)^\beta \cdot \exp\left(\gamma \frac{\Delta - 1}{\Delta + 1}\right). \quad (\text{S3})$$

Proof. On every edge uv we have $1 \leq d_u, d_v \leq \Delta$, hence $d_u d_v \leq \Delta^2$ and $d_u + d_v \leq 2\Delta$, and the normalised imbalance $\text{imb}(x, y) = |x - y|/(x + y)$ is maximised over $\{(i, j) : 1 \leq i \leq j \leq \Delta\}$ at $(1, \Delta)$, so $|d_u - d_v|/(d_u + d_v) \leq (\Delta - 1)/(\Delta + 1)$. Since $\alpha, \beta, \gamma \geq 0$, the maps $t \mapsto t^\alpha$, $t \mapsto t^\beta$ and $t \mapsto e^{\gamma t}$ are non-decreasing, so $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \Delta^{2\alpha} (2\Delta)^\beta \exp(\gamma(\Delta - 1)/(\Delta + 1))$, and summing over the m edges yields (S3). For $\gamma > 0$ and $\Delta \geq 2$ the bound is attained if and only if $\alpha = \beta = 0$ and every edge has degree pair $(1, \Delta)$ (for connected G , $G \cong K_{1, \Delta}$). If $\alpha > 0$ or $\beta > 0$ the three factor-wise maxima are not simultaneously attainable and the inequality is strict. ■

Proposition 5. *Let G be a connected graph of size m and minimum degree*

δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,

$$LO(G; \alpha, \beta, \gamma) \geq m \cdot \delta^{2\alpha} \cdot (2\delta)^\beta, \quad (\text{S4})$$

with equality if and only if G is δ -regular, except in the degenerate case $(\alpha, \beta, \gamma) = (0, 0, 0)$, where $LO(G; \alpha, \beta, \gamma) \equiv m$ and the bound holds with equality for every graph.

Proof. On every edge $d_u, d_v \geq \delta$, so for $\alpha, \beta \geq 0$, $(d_u d_v)^\alpha \geq \delta^{2\alpha}$ and $(d_u + d_v)^\beta \geq (2\delta)^\beta$. For $\gamma \geq 0$ the exponential factor is at least 1 because the imbalance is non-negative, and multiplying and summing over the m edges yields (S4). If $(\alpha, \beta, \gamma) = (0, 0, 0)$ then $LO(G; 0, 0, 0) = m$ equals the bound for every graph, while otherwise equality requires equality in every active per-edge factor, which for $\alpha > 0$ or $\beta > 0$ forces $d_u = d_v = \delta$ on every edge. If $\alpha = \beta = 0$ and $\gamma > 0$ it forces $d_u = d_v$ on every edge, so the connected graph G is regular, hence δ -regular. Conversely, on a δ -regular graph every edge has degree pair (δ, δ) and equality holds. ■

The next proposition gives a single uniform template for bounding $LO(G; \alpha, \beta, \gamma)$ both above and below. The bound is in terms of any classical bond-incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

Proposition 6. *Let G be a simple graph with maximum degree at most Δ . Let $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$ be a BID index with edge contribution $h(i, j)$ strictly positive on every degree pair (i, j) that appears in G . Define*

$$c_{\min}^h = \min_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)}, \quad c_{\max}^h = \max_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)},$$

where the extrema range over the degree pairs actually appearing in G . Then

$$c_{\min}^h \cdot I_h(G) \leq LO(G; \alpha, \beta, \gamma) \leq c_{\max}^h \cdot I_h(G), \quad (\text{S5})$$

with equality on either side if and only if every edge of G attains the corresponding extremum of ψ/h .

Proof. On every edge uv we have $h(d_u, d_v) > 0$ and $c_{\min}^h \leq \psi(d_u, d_v; \alpha, \beta, \gamma)/h(d_u, d_v) \leq c_{\max}^h$. Multiplying by $h(d_u, d_v)$ and summing

over $E(G)$, with $\sum_{uv} h(d_u, d_v) = I_h(G)$ and $\sum_{uv} \psi = LO(G; \alpha, \beta, \gamma)$, gives (S5). Each side is an equality if and only if every edge attains the corresponding extremum of ψ/h over the appearing pairs. ■

Proposition 6 is a *sharp graph-specific* comparison, whose constants depend on the degree pairs that actually appear in G , and a uniform version follows immediately by enlarging the range of the extrema.

Corollary 7 (Uniform Δ -bounded version). *Assume the hypotheses of Proposition 6, but with h strictly positive on all of $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ and the extrema taken over P_Δ . Then inequality (S5) holds simultaneously for every graph of maximum degree at most Δ , with constants independent of the individual graph. Equality on either side holds if and only if every edge of G realises a degree pair attaining the corresponding extremum over P_Δ . The uniform constants are in general strictly worse than the graph-specific ones, since the extrema are taken over a superset.*

S10 Data quality and open provenance items

Every value in Table 4 of the main paper was traced to a source (per-value record in the repository). The items that remain open are listed here so that they can be read in one place.

- (i) *Thirteen gas-phase entropies have no $S^\circ(\text{gas})$ entry in the NIST WebBook.* Seven of them (2,3-, 2,4-, 2,5- and 3,3-dimethylhexane, 3-ethylhexane, 2,2,3- and 2,2,4-trimethylpentane) agree, within tolerance or as documented revisions between editions, with the 1947 API Research Project 44 tables or a compiled octane dataset. The remaining six (4-methylheptane, 2,2-dimethylhexane, 3,4-dimethylhexane, 3-ethyl-2-methylpentane, 3-ethyl-3-methylpentane, 2,3,3-trimethylpentane) could not be matched to any primary source. The 1947 API Research Project 44 tables [12] list gas-phase entropies for all six. These differ from the values used by $0.07\text{--}2.2 \text{ cal mol}^{-1} \text{ K}^{-1}$, as that edition does for isomers whose later revisions we could confirm. No later primary determination was located, so the values used are compiled-dataset values. Substituting

the 1947 values changes the S correlations by at most 0.018 in $|r|$ and puts M_1 ahead of HM by 0.004.

- (ii) *2,3,3-trimethylpentane* S . The value 101.31 found in our earlier compilation is a copy of the 2,2,3-trimethylpentane entry and is not used. The value used, 102.06, is that of the compiled 18-octane dataset reproduced by Mondal et al. [14] (the one-decimal compilation of Ediz [13] gives 102.10). This change moves the S correlations by at most 0.009, and the entry used remains a secondary value (item (i)).
- (iii) *Four entropies differ from the compiled octane dataset of Ediz [13]* (octane 111.55 vs 111.70; 2,2-dimethylhexane 103.13 vs 103.40; 2,2,4-trimethylpentane 101.81 vs 104.10; 2,3,3-trimethylpentane 102.06 vs 102.10). For octane the value used agrees with the NIST WebBook (111.63). Substituting the compilation values changes all twelve S correlations, by up to 0.043 in $|r|$, and moves the margin between HM and M_1 at the top from 0.001 to 0.007.
- (iv) *2,2,3,3-tetramethylbutane* ΔH_{vap} is a benchmark-compilation value (8.41) whose reference state is unresolved. The compound is a solid at 298 K, and NIST lists sublimation and standard vaporization enthalpies near 10.3 kcal mol⁻¹. We repeated the matched-budget comparison of Supplementary Section S3 without this isomer, and the vaporization-enthalpy gain in the $(-2, 2)$ box persists (93/100 and 96/100 seeds instead of 100/100). The ΔH_{vap} column winner of Table 5 of the main paper is unchanged, and no conclusion of the paper rests on this value.
- (v) *Acentric factors* of 2,2,4-trimethylpentane and 2,2,3,3-tetramethylbutane differ from an alternative compilation by 0.002–0.004, and these alternative values change several ω correlations at the third decimal and leave the ω ranking unchanged.

None of these items moves a correlation of Table 5 of the main paper by more than 0.043. The only bolded entry at stake is the S winner, where HM leads M_1 by 0.001 with the values used. The 1947 substitution of item (i) reverses that order by 0.004, a margin without meaning at this resolution. None of the items enters the pre-registered nonane and

boiling-point tests of Sections 5.3 and 5.4 of the main paper. Those tests use no entropy, vaporization-enthalpy or acentric-factor value of Table 4 of the main paper, and the pooled boiling-point test re-extracts the octane boiling points under its own rule. Item (ii) was a transcription error in our earlier data file, corrected here, while the remaining items are limitations of the compiled octane benchmark rather than of the analysis.

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